THE 11TH INTERNATIONAL ANKLE SYMPOSIUM

SEOUL 2026
INTERNATIONAL ANKLE SYMPOSIUM

Foot Core vs Trunk Core Training in CAI

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One sprain. Then what?

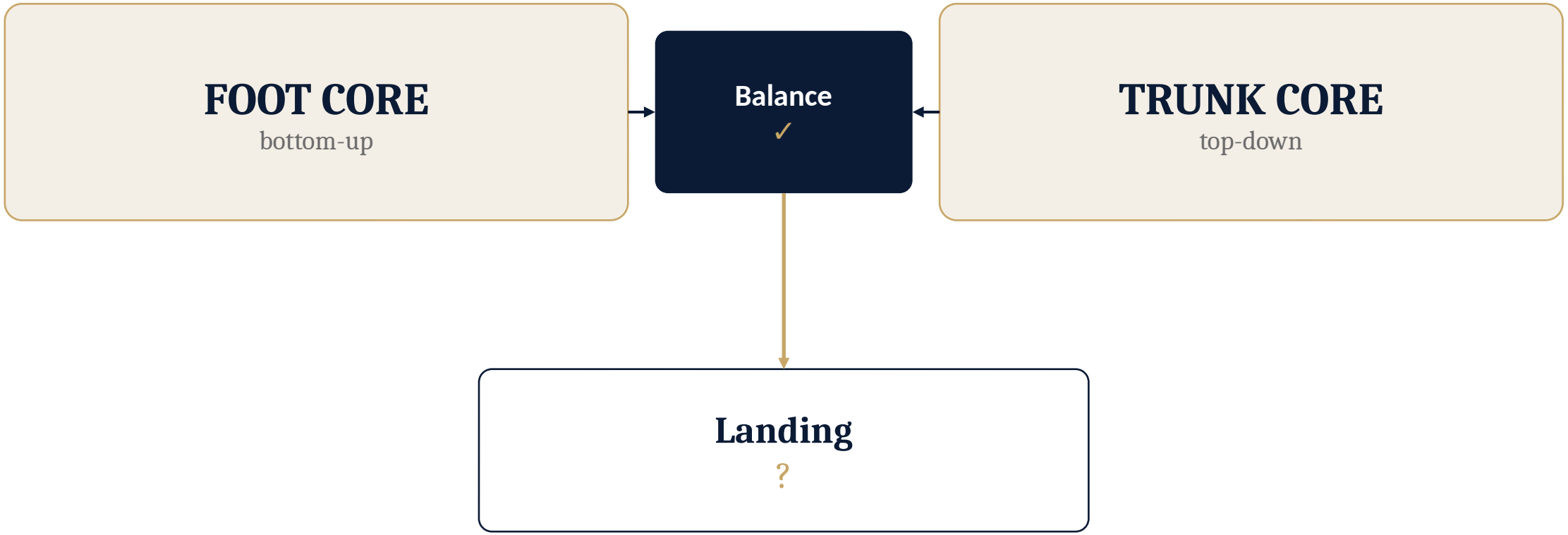
40%

of first-time ankle sprains become chronic instability

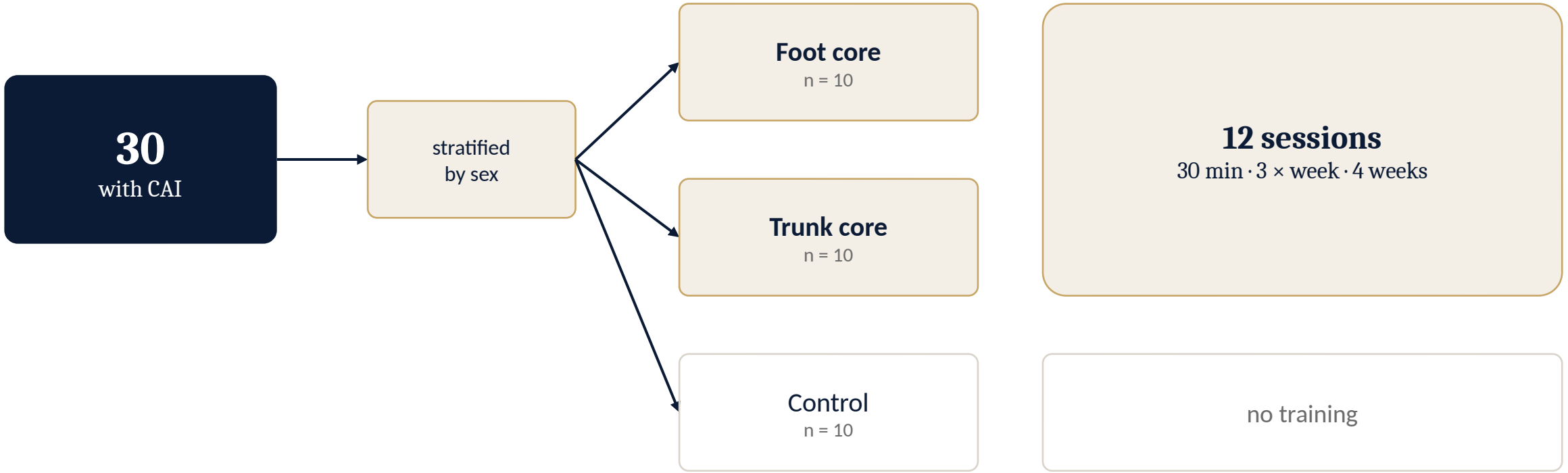
Not pain. Giving way — at landing.

Hertel & Corbett, J Athl Train 2019

Foot, or trunk?



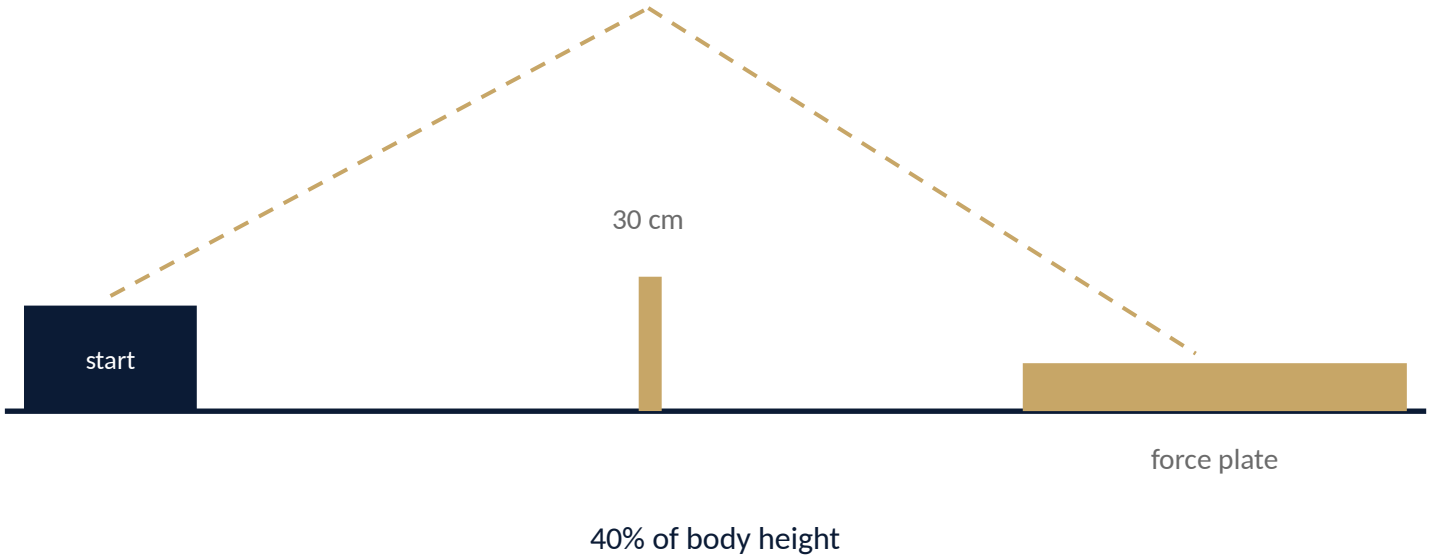
Thirty people. Three groups. Four weeks.



pre-test → 4 weeks → post-test

The task

jump · clear · land on one leg · hold 10 s



COP
static stance

DPSI
force plate

3D kinematics
ankle · knee · hip · trunk

0–200 ms · SPM

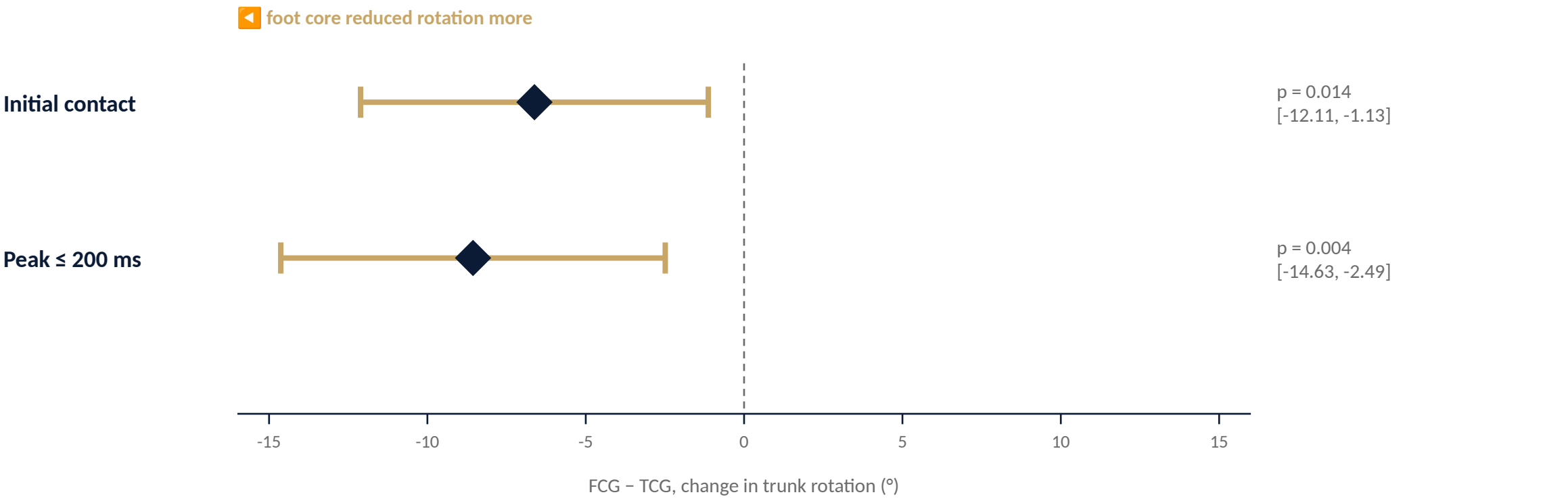
Balance: nothing moved

0

interactions
across every balance measure

COP velocity	95% ellipse	DPSI

Trunk rotation: foot core beat trunk core



interaction IC p = 0.016 · peak p = 0.004 · FCG vs CG at peak p = 0.053 · ankle · knee · hip: none

...and across the whole landing window



foot core < trunk core 25–100% p = 0.010

foot core < control 80–100% p = 0.015

Interaction across the entire window, p = 0.010

Figure 1 · SPM interaction
Figure 2 · pairwise change waveforms
(from manuscript)

Train the first link. The last link quiets.



Better initial contact → less to correct upstream

limits n = 10 per group · 4 weeks · no follow-up

Take-home

Add foot core early.

Alongside balance and strength work — not instead of it.

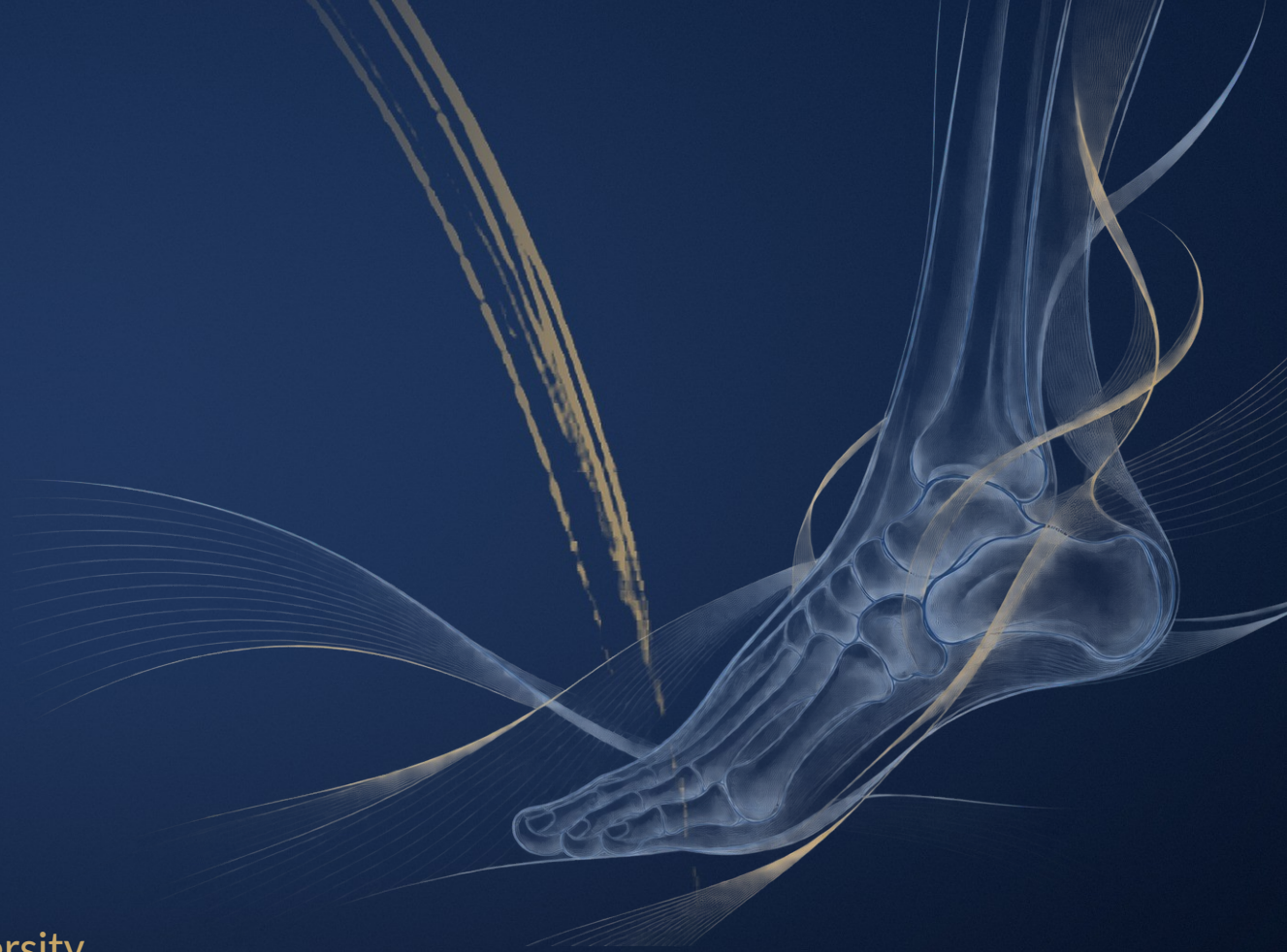
4 weeks

12 sessions

30 minutes

Thank You

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